

Replacing Bak Choy with Cabbage: Alternative Processes and Scenarios and Inspired Creativity in Documenting Intangible Forms of Culinary Culture Heritage

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Abstract

Unlike physical artifacts, Intangible forms of Cultural Heritage (ICH) like culinary processes rely on community transmission and often improvised methods of creation, making it difficult to preserve in digital form. To design a process for documenting culinary ICH of Chinese cuisine, we prototyped a workshop for ICH-knowledgeable practitioners to visualize their culinary processes and alternative scenarios like missing ingredients, use of different instruments, and deviations from traditional recipes. Using text-to-image tools to facilitate this process, we found that although GenAI misrepresented traditional cooking methodologies and materials, it could facilitate reflection on cultural memories and considerations of alternative methodologies. This leads them to design creative outcomes that transcend normative forms of ICH documentation, imagining dishes that fit the idea of recorded ICH while differing in precise form. This work shows how GenAI-supported co-creation can help document contingencies in ICH processes.

CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

Keywords

Chinese Cuisine, Intangible Culture Heritage, Human Food Interaction, Generative AI

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1 Introduction

Safeguarding intangible cultural heritage (ICH) involves the processes of research, preservation, promotion, and enhancement [89].

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Although documentation is an important aspect of safeguarding, it remains challenging because ICH often depends on oral traditions, embodied practices, and tacit or intuitive knowledge [50]. Moreover, ICH has been understood as an evolving process rather than a fixed entity, continuously shifting in response to social and historical changes [9]. Therefore, each form of ICH requires individual approaches to its documentation, transmission, and preservation that account for its inherent flexibility.

Culinary traditions, as one of the forms of (ICH) recognized by UNESCO, present particular challenges for safeguarding and documentation due to their regional diversity, distinctive techniques, and the cultural meanings embedded within recipes and cooking methods [31, 59]. The act of cooking is inherently interpretative and adaptive, shaped by environmental and social contingencies. The context-specific nature of cooking practices, the complexity of ingredients and the depth of historical development, make ICH cuisine recipes difficult to accurately record [26, 31, 93].

Within this context, although textual and photographic records of cooking processes are valuable, they are often insufficient for sustaining or transmitting the full experiential and situational dimensions of culinary heritage. Even when a complete recipe is documented, its realization remains contingent on multiple factors such as the availability of ingredients, the tools and utensils used, and the preferences or circumstances of those who prepare and consume the food [23, 72]. Therefore, ICH cuisine should be documented in ways that acknowledge its dynamic, adaptive, and interpretative nature.

New technologies such as AI have increasingly demonstrated their potential to support the safeguarding of cultural heritage [34, 43]. Beyond heritage preservation, AI has also contributed to Human–Food Interaction (HFI), where it has been applied to the documentation, design, and envisioning of food practices [17, 76]. These developments reveal new possibilities for recipe generation, food promotion, and the reinterpretation of culinary practices. Following this trajectory, recent study has shown that generative AI (GenAI) can further support Chinese ICH cuisine practitioners in recreating and refining traditional recipes [66].

Building on these premises, we designed a study consisting of a series of in-person workshops to explore how ICH cuisine can be documented adaptively through the support of GenAI. The workshops involved chefs specializing in Chinese ICH cuisine, with GenAI employed to assist practitioners in documenting their culinary practices and experimenting with recipes and techniques that adapt to different contexts. Through these sessions, participants

co-created visual representations of their dishes using GenAI, providing insights into the challenges of translating tacit knowledge into prompts, the influence of GenAI on their perceptions of ICH, and their evaluations of the generated outcomes. This approach seeks to bridge the gap between tradition and innovation, contributing to the preservation and sustainable development of ICH cuisine in contemporary contexts.

This study is guided by three research questions:

RQ1: How can the diversity of ICH cuisine be effectively documented in ways that adapt to contingencies and contexts?

RQ2: How does GenAI impact perception and understanding of practitioners towards the ICH cuisine?

RQ3: What kind of challenges do the practitioners meet when working with GenAI to help document and enhance ICH cuisine?

According to our findings, GenAI can assist in documenting ICH culinary practices across different scenarios by generating visual and textual outputs, informed by practitioners' cooking processes and reflections. However, using GenAI for culinary ICH documentation also presents several challenges: (1) Transforming non-verbal and intuitive cooking processes into text-based prompts risks omitting important details of traditional techniques. (2) Cultural biases embedded in GenAI's training data may lead to misrepresentations of non-Western culinary practices. (3) Participants' individual understandings of their own cultural practices and their differing evaluations of GenAI outputs in terms of accuracy and authenticity further complicate this process. If these challenges are carefully addressed in future research and system design, GenAI could more effectively support the documentation of recipes, offering alternative and adaptive ways to preserve and reinterpret ICH cuisine.

2 Background

2.1 Culinary Intangible Cultural Heritage

Food memories are deeply connected to personal experiences, communal culture, and emotional ties [59]. Traditional dishes are often linked to local celebrations and social gatherings [93, 94], reinforcing regional cultural identities [92]. The documentation and promotion of traditional cuisines also contribute to sustainable development by supporting local agriculture, tourism, and economic growth [91]. These cultural impacts collectively contribute to their recognition as intangible cultural heritage of humanity.

After UNESCO first inscribed food-related culture on the List of Intangible Cultural in 2010, the concept of food heritage has garnered increasing attention in both academic and societal contexts [44, 80, 90]. Currently, nearly 40 culinary traditions are recognized on the UNESCO list [1].

The approach to preserve regional food traditions typically involve cataloging indigenous knowledge and culinary practices [93]. However, most ICH food consists of traditional dishes using family recipes that vary from one to another, and the transmission of culinary knowledge and techniques often relies on oral tradition or handwritten notes [31, 93]. Furthermore, some traditional foods need to be adapted to align with changes in modern society [23, 72], such as the dwindling availability of endangered materials [79], traditional cookers being replaced by evolving kitchen tools [86], the decreasing size of families [78], and shifting health standards

[11, 88]. It causes significant challenges to documentation of ICH culinary recipes in dynamic and adaptive way [26, 31].

2.2 Digital ICH

A wide range of digital resources have been developed to make ICH more publicly accessible and engaging [4, 4, 97]. Social media has emerged as another important medium for the safeguarding of ICH. HCI researches shows that livestreaming can increase visibility and community engagement around ICH practices [67–69], while short-video platforms serve as spaces for creative exchange and reinterpretation between practitioners and viewers [96].

At the same time, the advancement of Information and Communication Technologies (ICT) and computational heritage has significantly contributed to the preservation and transmission of Intangible Cultural Heritage (ICH) [45, 65]. For instance, immersive digital reconstructions such as web-based and VR-enabled 360-degree experiences allow users to navigate heritage sites virtually [22, 82], while other VR and AR as well as MR applications expand the visual reach and interpretation of cultural artifacts [7, 36, 46, 62]. Other interactive approaches, including tangible user interfaces, have been designed to capture and sustain indigenous forms of knowledge [27]. Serious and educational games have been adopted to enhance public awareness and appreciation of both tangible and intangible heritage [37, 77].

As a large language model (LLM), GenAI has made significant advancements in recent years, enabling the generation of visual depictions from even rough text notes [49, 74, 81, 99]. There are studies that have explored collaboration with GenAI tools through text-to-image generation, which aids in creative support and the preservation of unclear or vague materials [30, 75, 95]. Recent work on GenAI extends this digital trajectory by supporting creative and cultural practices across multiple domains. Researchers have explored its role in writing [15, 100], co-artistic creation [38, 42], debate [102], and design [60, 64, 71]. In cultural heritage contexts, GenAI has been adopted to co-create, reconstruct, and visualize artifacts and exhibitions [34, 51–54], and to support embodied understandings of ICH through movement and visual expression [43]. Further studies explore how GenAI captures internet-based cultures [106] and enriches interactive narratives and gaming environments [32, 64, 71, 87, 105, 107–109], allowing audiences to encounter imaginative cultural representations.

2.3 Human Food Interaction

Recent advances in GenAI have expanded the ways ICH can be visualized, reconstructed, and shared, which also open new opportunities to engage with culinary heritage. This intersection between GenAI-supported cultural documentation and food-centered culture leads naturally toward the field of Human–Food Interaction (HFI).

The growing academic interest in food and taste within the HCI community has given rise to the concept of HFI [8, 17]. Researchers in this field conduct design experiments and participatory workshops to explore everyday eating practices [16, 18, 35] and to imagine sustainable futures for food culture [24, 63]. Technological interventions play a central role in these explorations. Immersive technologies such as VR, AR, and metaverse environments have

been used to construct multisensory food experiences and speculative dining scenarios [20], while AI contributes to novel approaches in computational gastronomy and food design [3, 39, 48].

Designing with and around food also encourages playful, aesthetic, and social engagement [5]. Digital creators employ games to record and disseminate culinary heritage [21], and interactive food exhibitions increasingly blend art, design, and industry innovation [47]. Other themes within HFI include computational dining experiences [25], food journaling for reflection and wellbeing [70], and the influence of online food content creators on social and cultural practices [98].

Despite there are shared interests of ICH and HFI in technology supported food cultural expression, research that bridges these two domains remains scarce. This project seeks to fill that gap by employing GenAI as a participatory tool for documenting and reimagining culinary ICH practices in alternative scenerios, connecting the technological mediation of cultural heritage with the experiential richness of Human–Food Interaction.

3 Methods

To explore how practitioners can document ICH culinary processes in dynamic and adaptive ways, we organized a series of in-person workshops that encouraged them to imagine alternative cooking scenarios through speculative design methods supported by GenAI [6, 19]. At the onset of each workshop, we interviewed participants regarding their traditional cooking methods and the specific details of their chosen ICH cuisine. We then introduced GenAI as the tool for visualization and invited participants to engage with it for the purpose of documenting, representing their alternative forms of the ICH dishes.

For those participants who had limited access or prior experience with GenAI, our research team provided hands-on assistance in navigating the interfaces. This support enabled participants to concentrate on the creative elements and the expression of their culinary ideas. Throughout the workshops, participants were prompted to employ the think-aloud protocol to verbalize their thoughts and emotions, which were audio-recorded. Upon the completion of each workshop, we conducted follow-up semi-structured interviews with the participants to further explore their experiences.

We selected ChatGPT integrated with Dall-E3 as our primary text-to-image generation tool due to its advanced natural language understanding capabilities and user-friendly interface. In contrast to Midjourney or Stable Diffusion, which often demand proficiency in writing detailed prompts, ChatGPT allows participants to simply input their intentions, making it accessible even to those without prior GenAI knowledge. Additionally, at the request of some participants, we also incorporated Doubao and Zhipu into the process.

3.1 Participants

Fifteen participants, ranging in age from 26 to 60 years with an average age of 37 years old, were enlisted for this study. All participants were Chinese citizens with a minimum of six years of experience in preparing ICH Chinese cuisines. Among them, three were professional chefs. The detailed demographic information of the participants is presented in Table1.

The criteria for selecting Intangible Cultural Heritage (ICH) cuisines were based on the official list of ICH - certified food items recognized by relevant Chinese ICH departments at all levels, including provincial, municipal, and prefectural ones. This selection was made because these cuisines, with their authenticity and cultural significance, are ideal for exploring the digital documentation of ICH culinary processes[2, 73].

3.2 Procedure

Initially, we conducted interviews with the participants, covering their cooking backgrounds, the specific ICH dish they had selected for the workshop, and other pertinent details. We then delved deeper into the dish’s history, variations, and unique characteristics. These variations encompassed aspects such as different target audiences (tasters), ingredient substitutions, the influence of the cook’s style, recipe modifications, temporal and spatial factors, cooking utensils, and even common mistakes. This speculative and comprehensive exploration of the ICH cuisine’s variations aimed to facilitate a multidimensional approach to its documentation [26, 31, 57]. The participants were also invited to cook dishes. During the cooking process, The researchers also recorded videos and took detailed notes throughout the process to capture as much critical information as possible.

Next, we provided a practical demonstration of how to use the GenAI interface (specifically, ChatGPT) to generate both textual and visual content through interactive dialogues. After a period of hands-on practice, participants were guided to generate AI-assisted images and recipe, drawing inspiration from the cuisine itself and the insights gleaned from the earlier interviews about its variations. For instance, they were encouraged to envision how the dish would transform if alternative cooking methods were employed, as suggested by the AI. Additionally, participants were given the freedom to innovate the ICH cuisine according to their personal creativity and preferences. After each instance of AI-generated image creation, we conducted interviews with the participants using the photo-elicitation method [40, 41].

Finally, we conducted semi-structured interviews with the participants to gather their feedback on various aspects, including their emotional responses to the AI-generated content, the overall cooking and interaction processes, the knowledge gained, memories formed, and their behavior and experiences in using AI.

3.3 Data Collection and Analysis

Throughout the interview and interaction processes between the participants and the AI, we recorded audio and took detailed field notes. Additionally, the transcripts of their interactions with the AI were also captured. Participants were informed about the purpose of the study and gave their consent for their cooking processes, interviews, and GenAI-generated materials to be recorded and analyzed for research purposes.

These data were initially transcribed and then independently open-coded by four researchers following the principles of thematic analysis [10]. Multiple rounds of iterative discussions and coding were conducted until a consensus was reached among the researchers regarding the final themes. This process led to the organization of the interview data into several themes: GenAI for

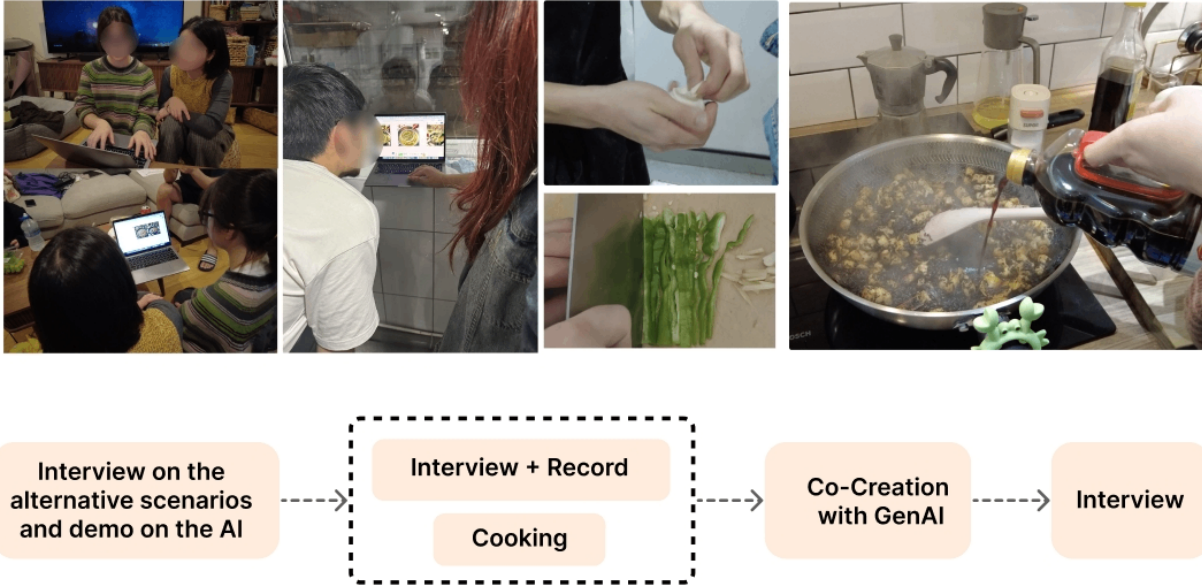


Figure 1: The workflow of the workshops

Table 1: Demographic Information of Participants

ID	Dish	Gender	Age	Occupation	Experience of Cooking
P1	Hakka-style Stuffed Tofu	Male	29	Amateur	10 years of Hakka cuisine
P2	Claypot Rice	Male	56	Chef	30 years of Cantonese cuisine
P3	Basil Stir-fried Clams	Male	26	Amateur	6 years of Cantonese cuisine
P4	Hunan Style Stir-fried Beef	Male	50	Chef	30 years of Hunan cuisine
P5	Garlic Vermicelli Shrimp Pot	Female	30	Amateur	10 years of Sichuan and Chongqing cuisine
P6	Juicy Soup Buns	Male	27	Chef	7 years of Dim Sum
P7	Liuyang Steamed Pork Ribs	Female	25	Amateur	6 years of Jiangxi Cuisine
P8	Yu-Shiang Shredded Pork	Male	25	Amateur	6 years of Henan cuisine
P9	Stewed Pork with Preserved Vegetables	Female	27	Amateur	6 years of Jiangsu and Zhejiang cuisine
P10	Guiyang Fermented Soybean Hot Pot	Male	43	Chef	10 years of Sichuan cuisine, Guizhou cuisine
P11	Shaanxi Oil-spilled Noodles	Female	32	Chef	8 years of Shaanxi cuisine, Fusion French Cuisine
P12	Wonton Noodles	Female	31	Amateur	10 years of Cantonese cuisine
P13	Zhaoqing Zongzi	Female	58	Amateur	30 years of Cantonese cuisine
P14	Chaoshan Beef Balls	Male	40	Chef	20 years of Chaoshan cuisine
P15	Hakka Salt-baked Chicken	Male	55	Chef	30 years of Cantonese cuisine

Recording ICH Cuisine, GenAI for Innovating the ICH cuisine and Perceived Misrepresentation from GenAI.

4 Results

4.1 Documenting Alternative Scenarios of ICH Culinary Practices

In this section, we summarize the alternative scenarios documented by the participants and how they did during the engagement. Through these practices, participants' engagements with GenAI provided

valuable insights into the ICH cuisine in various forms. The Images and prompts were presented in Fig.2

4.1.1 Missing Ingredients. Several participants highlighted challenges related to missing or unavailable ingredients when cooking outside of their native country.

P4, a chef with 30 years of experience in Hunan cuisine working at a restaurant in London, shared that the restaurant struggles to find the same variety of chili peppers commonly used in China. To address this, they substitute the Chinese spiral chili peppers with Turkish chili peppers to maintain freshness and reduce costs. Similarly, P5, a food amateur who has lived abroad for 15 years, explained that typical foreign supermarkets usually don't carry large prawns with heads. As a result, when she uses the more commonly available peeled shrimp, the flavor is not as robust as when using whole prawns. P9, a chef working at a Burmese restaurant, noted that the pork belly available abroad is typically cut into thinner strips, unlike the thicker cuts used in China, leading to smaller pieces of pork belly being used. This, in turn, changes the texture and flavor of the dish. Overall, P4, P5, and P9 all emphasized that the inability to find specific ingredients abroad significantly affects the flavor of the dishes they prepare. P1 also supported this observation, mentioning that when making Hakka-style Stuffed Tofu, the absence of starch for thickening the sauce would result in a less tender and smooth texture. P9 further noted that omitting the aromatic leaves typically grilled on a hot iron plate in their restaurant would negatively affect the dish's flavor.

4.1.2 Cooking for Others. Many participants discussed how they adjust their cooking techniques and recipes when preparing food for different audiences. P4 explained that since the restaurant serves many foreign customers, dishes like black pepper beef often need to be adapted to suit local tastes. To standardize the presentation and ensure consistency in the dishes, he cuts all beef slices to the same thickness, even for dishes like Hunan Style Stir-fried Beef, which originally uses thin, square beef slices in the traditional Hunan style. When preparing this dish for foreign customers, P4 reduces the amount of chili to better accommodate their palates. P5 also shared her experience of adjusting recipes when cooking for foreign friends, such as making Garlic Vermicelli Shrimp Pot by replacing cilantro with parsley, using butter instead of soy sauce for the garlic sauce, and adding onions for extra flavor. Additionally, P10 noted that regional preferences also influence how he adjusts dishes. For example, when cooking a Guiyang Fermented Soybean Hot Pot for people from Guangdong, he reduces the amount of chili to suit their milder taste preferences. These examples demonstrate how chefs and home cooks modify dishes to fit the tastes and preferences of different customers or audiences.

4.1.3 Making Mistakes. Several participants shared how errors during the cooking process can lead to suboptimal results, highlighting the role of GenAI in presenting the mistakes for other people. P9 mentioned that she usually uses brown sugar instead of rock sugar when making caramel, as rock sugar tends to burn easily. P4 also noted that when the heat is too high while stir-frying beef, the meat becomes tough and the bottom of the pan can burn. P3 commented that if the oil temperature is too high when frying minced garlic, it can cause the garlic to burn. P5 explained that adding garlic to

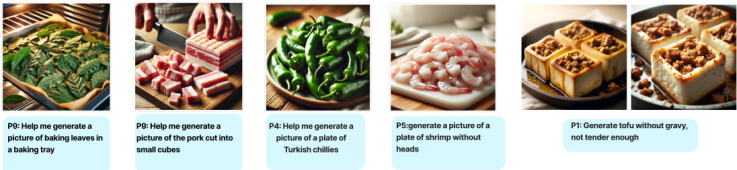
the oil before it has reached the correct temperature results in less aromatic garlic. P9 further emphasized that garlic should be added just before the dish is finished cooking to release its full fragrance. P5 also noted that when preparing Garlic Vermicelli Shrimp Pot, if the vermicelli is soaked for too long, it becomes too soft, and the temperature of the water used to soak the vermicelli can also impact the texture. P6 explained that when preparing dough, if it is not covered properly, it will dry out and crack, affecting the texture of the dish. Additionally, P9 mentioned that when making Stewed Pork with Preserved Vegetables, failing to create a well in the ingredients for the braising sauce will result in uneven seasoning, leading to a less flavorful dish. In response to these cooking mistakes, P2 pointed out the usefulness of AI in visualizing these errors. He suggested that when teaching apprentices, AI could be employed to show the difference in outcomes based on varying heat levels, which would help them understand the effects of their mistakes. On the other hand, P5 expressed skepticism about AI's ability to accurately represent cooking errors, noting that humans typically avoid photographing failed dishes, and AI often lacks the data to illustrate common mistakes, such as overly wet dough or failed fermentation in bun-making.

4.1.4 Variations of Dishes. Several participants shared their experiences with creating variations of traditional dishes, whether by adjusting the flavor profile, altering ingredients, or experimenting with new preparations.

P7, for example, described an interesting variation of Liuyang Steamed Pork Ribs by using dried chili powder instead of the usual fresh chili peppers. This simple swap resulted in a completely different flavor profile, where the heat from the dried chili added a smoky and earthy depth to the dish. To visualize this alternative approach, P7 generated an image that showcased the dish with dried chili powder, demonstrating how such a substitution could be integrated into the recipe while maintaining the dish's integrity. P13 also explored variations in Zhaoqing Zongzi by experimenting with different fillings. Typically, alkaline Zongzi are made without any filling, offering a plain but chewy rice texture. However, P13's household also enjoys sweet Zongzi made with red bean paste, offering a contrasting sweet filling to balance the savory nature of the traditional Zongzi. This variation not only showcases the versatility of Zongzi but also reflects regional preferences for different fillings, such as savory or sweet, which can vary significantly across China. Similarly, P8 demonstrated how a Yu-Shiang Shredded Pork dish could be modified to cater to different preferences. By adjusting the amount of pork in the recipe, P8 created two versions of the dish—one with more meat and one with less. These variations were designed to accommodate different dietary preferences or simply to offer a choice between a richer, meat-heavy dish and a lighter, more vegetable-forward option. The visual representation of these two versions of the dish allowed for a deeper understanding of how ingredient proportions can alter the overall eating experience.

4.1.5 Using Different Instruments. The choice of cooking instruments plays a crucial role in shaping the final flavor and texture of dishes. Several participants discussed how different instruments and cooking methods, whether historical, traditional, or modern, affect the outcome of their cooking.

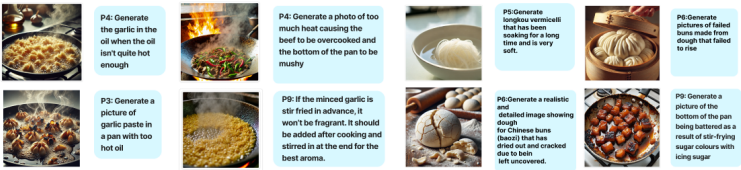
1.Missing Ingredients



2. Cooking for Others



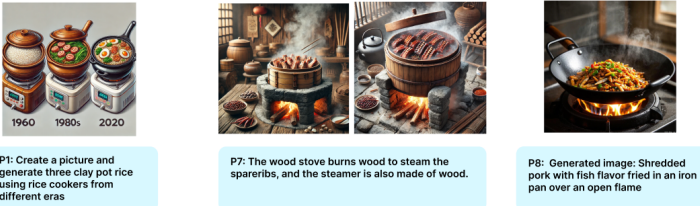
3.Making Mistakes



4.Variations of Dishes



5.Using Different Instruments



6.Different Workflows

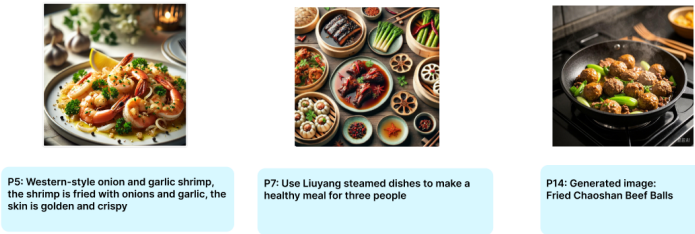


Figure 2: The Images and prompts of the participants documenting the alternative scenerio of the ICH dished.

P2, for instance, pointed out the influence of different rice pots used over the years to cook Claypot Rice. The cooking pots have evolved, and with these changes, the texture and flavor of the rice have been affected. P2 generated images of rice pots from different

eras, visually highlighting the impact that the choice of cooking vessel can have on the dish. P7, on the other hand, described how the use of a wood-fired stove and wooden steaming baskets for steaming pork ribs can result in a significantly enhanced flavor.

According to P7, the aroma from burning wood infuses the food during the cooking process, creating a unique fragrance that cannot be replicated by modern gas stoves or electric steamers. Similarly, P8 shared his preference for using a cast-iron wok when cooking Yu-Shiang Shredded Pork. He emphasized that the high heat achievable with an iron wok is key to achieving the authentic "wok hei", or the smoky charred flavor, that defines many stir-fry dishes.

4.1.6 Different Workflows. During this process, some participants also explored the cooking of dishes under different workflows. These include alternative scenarios featuring different styles, preparation techniques, or serving various purposes. For instance, P7 explored flavor variations by substituting dried chili powder when steaming pork ribs. She explained, *"When I steam the ribs, replacing the usual ingredients with dried chili powder gives it a completely different flavor."* This adjustment not only changed the dish's taste but also illustrated how participants personalize traditional recipes through simple ingredient substitutions to create entirely new flavor experiences.

On the other hand, P8 highlighted regional adaptations by contrasting their current preparation of Yu-Shiang Shredded Pork with the Zigong version. She noted, *"The Zigong version of the dish uses more oil and salt, with fewer ingredients, primarily scallions. So I tried the Zigong version of Yu-Shiang Shredded Pork to reflect this."* This regional comparison not only preserved the core characteristics of the dish but also adapted to varying regional taste preferences, showcasing the diverse evolution of traditional dishes across different cultural backgrounds.

Additionally, P13 experimented with traditional and sweet variations of zongzi, the Chinese rice dumplings. She remarked, *"We usually make alkaline water zongzi, which contain no fillings. At home, we also prepare sweet zongzi with red bean fillings."* This innovation maintained the fundamental form of traditional zongzi while introducing new flavor elements, catering to different dietary preferences and enhancing the dish's diversity and adaptability.

4.2 Perceived Misrepresentation of the Generative Process

During the process

4.2.1 Misrepresentation of the Ingredients. : The participants identifying inaccuracies in how ingredients were represented in GenAI generated images compared to their expectations or knowledge of traditional practices. This led to discussions about how GenAI might better capture the intended culinary details.

For example, during an interview, P3 said:

"These are indeed larger shells, but the presence of chili peppers in the image is not quite right."

While P3 mentioned that the size of the shell matched their prompt, they expressed dissatisfaction with the inclusion of chili peppers, which did not align with the traditional recipe. Similarly, P4 commented on the incorrect preparation state of chili peppers:

"The peppers in the generated image are not in the right state. They should be chopped, not whole peppers." (Fig.3)

4.2.2 Inaccuracy in Generating representation of utensils. : Participants also identified inaccuracies in the representation of tools within GenAI generated images, highlighting a mismatch between



Figure 3: P4 mentioned these images failed to capture the intended culinary details

the depicted tools and those traditionally used in specific culinary practices. These discrepancies often stemmed from cultural biases embedded in GenAI's training datasets. For instance, during an interview, P6 remarked:

"Its pasta pressing machine is also Western style, it is the kind used for French cuisine."

This observation emphasized that the tool depicted in the image failed to align with the traditional equipment used for Chinese cooking. Participants noted that such mismatches could lead to a misrepresentation of cultural authenticity, particularly when documenting ICH practices. For tools like dough mixers or pasta presses, which are essential to many culinary workflows, these inaccuracies stood out as significant barriers to creating faithful visual representations.

To address these discrepancies, participants suggested refining prompts to include explicit descriptions of the tools and their cultural context. For example, specifying *"traditional Chinese dough mixer with a spiral hook"* instead of generic terms might help GenAI produce more accurate depictions. However, participants also acknowledged that such refinements might not fully resolve the issue, as the underlying training data often lacks diverse representations of non-Western culinary tools.

4.2.3 Misrepresentation the Stage of Cooking. : Another issue identified by participants was the misrepresentation of the stage of cooking process, showing the wrong status of the food in the GenAI generated images. Participants mentioned that these images often failed to capture the proper progression of the cooking processes, and appeared incomplete or inaccurate. For instance, P4 said:

"The peppers look like they haven't been cooked, they still appear very fresh. It feels more like different ingredients have just been assembled together."

The feedback highlights GenAI's difficulty in understanding the transformation of ingredients during cooking, such as softening, browning, or caramelizing. Participants expressed that these visual cues are crucial for representing the cuisine authentically and conveying the proper stages of the preparation. For example, the fresh appearance of the chili peppers contradicted the expected softened, cooked texture typically seen in the final dish.

In response to such inaccuracies, participants revised prompts to include more detailed descriptions of the desired ingredient states, such as *"softened peppers after cooking."* However, despite iterative



Figure 4: P4 mentioned these images failed to capture the proper progression of the cooking processes.



Figure 5: P5 expressed dissatisfaction with the overly polished or exaggerated styles.

refinements, the generated images often lacked the subtle transitions characteristic of traditional cooking processes, underscoring the limitations of GenAI in depicting realistic culinary states.

4.2.4 Unintended Aesthetic Style. : Participants also frequently mentioned the unintended and unrealistic aesthetic styles present in GenAI-generated images, highlighting that these images often detracted from the authenticity of the representations. The overly polished or exaggerated styles were particularly noticeable in the depiction of ingredients and cooking utensils. For example, P5 said:

“The style of the shrimp is a bit too cartoonish; it looks somewhat fake.” (Fig.5)

Similarly, P6, with a designer background, expressed dissatisfaction with the overall visual style. He emphasizing its artificial appearance:

“Since I studied design, I don’t like this overly greasy visual style—it’s obvious that it’s AI-generated. Look, the chopsticks in these images are all at the exact same 45-degree angle, and the folds on the buns look too fake.” (Figure 6) (Fig.6)

These comments reflect a broader critique of GenAI’s aesthetic bias, where images often appear overly stylized, rigid or lacking the normal imperfections of the real world. The uniformity in details, such as the identical angles of chopsticks or exaggerated textures of shrimp, further reinforces the artificial nature of the visuals, diminishing their cultural and practical relevance.



Figure 6: P6, with a designer background, expressed dissatisfaction with the overall visual style.

4.3 Engagement of the process when visualizing the alternative ICH cuisine

4.3.1 Increased Reflection of Cuisine-Making Details Through GenAI Images. Participants’ interactions with GenAI revealed its ability to prompt deeper reflection on the details of ICH cuisine-making processes, including gestures, toolkit, and cooking skills.

When using GenAI to record ICH cuisine, participants mentioned that some production techniques could not be expressed in words, so the subjects used gestures to describe them.

For example, during an interview, P6 said:

P6: I don’t even know how to describe the hand gesture for shaping buns, and even if I did, GPT probably wouldn’t understand. I think AI still has limitations when it comes to complex handmade techniques like folding pleats. It might be better to simplify it to just holding a bun in hand.

P2 also has similar expressions. He uses rich gestures as shown in the picture to express the use of ingredients.

The discrepancies between AI-generated tools and traditional ICH kitchen toolkit prompted discussions about cultural specificity.

P6: The dough hook generated here is incorrect; it generated a Western-style croissant dough hook. This type of hook makes the dough softer, but for making buns, we use this spiral-shaped dough hook. The inconsistency of GenAI in representing cooking skills, such as ingredient states or preparation methods, also prompted the participants to provide additional explanations to clarify these processes.

For example, P4 thought that the size of the peppers generated by AI was not right. P2 gave a more detailed reflection, pointing out that GenAI was inappropriate in generating clay pot rice, which would result in the rice not being loose. He then articulated the prompt to the AI to make the skills correct. P2: *“The rice was pressed with a spoon, but it was pressed too much, so it is not as loose as before, and the quality of the rice is not as good as before. It’s true. The air enters, but the heat is not enough, so it is forced out. The rice is not strong enough to withstand the heat, so it explodes when it is heated, and it will steam when there is enough steam.”*

P5: *"I thought of a small cooking tip: blanch the minced garlic first; otherwise, it's likely to burn and turn too dark when frying."*

4.3.2 Reviving Memories from the GenAI images. The Participants revived personal experiences and cultural stories related to ICH cuisine because of the content generated by GenAI. These visual outputs often served as memory triggers, helping participants reflect on their culinary journeys and rediscover techniques they had learned in the past.

For instance, P6 shared a detailed recollection of preparing a "poolish" starter for dough fermentation.

P6: *You're missing a step in this dough fermentation process—I usually prepare a "poolish" starter the night before. The poolish is made by mixing 100g of water, 100g of flour, and 1g of dry yeast, then leaving it in the fridge overnight. When kneading the dough the next day, I add the poolish in. This helps delay the aging of the dough and makes the steamed buns softer. I learned this method while working at a French pastry shop, combining Western and Chinese techniques.*

Similarly, P9 reflected on their mother's unique cooking method for frying dried mustard greens with lard, a technique passed down through family tradition. However, they noted the challenge of replicating this practice in London due to the unavailability of certain ingredients like lard, which led to subtle adaptations in their cooking.

P9: *"This method is still a bit different from how it's done back home because this dish was taught to me by my mother. When she stir-fried dried mustard greens, she would add some lard, but in London, I don't have the conditions to do that."*

P9 also recounted memories from their experience working at a Burmese restaurant, where dried bay leaves were roasted on an iron griddle in a method distinct from Chinese practices. These stories highlight how participants drew comparisons between their traditional cooking methods and the techniques encountered in new environments, often bridging cultural differences.

4.3.3 Acquisition of the Requisite Knowledge. During the workshops, participants expressed a need to acquire practical cooking knowledge tailored to their culinary practices, particularly when they encountered gaps in their understanding of traditional cooking techniques or specific ingredients. Some participants actively sought advice from GenAI regarding various dishes. For example, P10 recalled having eaten certain hot pot ingredients (such as egg dumplings) in the past but was uncertain about how to make them. He turned to AI for specific instructions, which helped him bridge the gap in his knowledge and enabled him to recreate the dish at home.

Similarly, P12 mentioned a particular wonton filling that a friend had once made, which left a lasting impression due to its delicious taste. However, she was unsure about the ingredients that would yield a similar effect. By consulting AI, she discovered suitable materials, such as century egg, that aligned with the flavor profile she desired. This interaction not only facilitated her understanding of the dish but also empowered her to experiment with her culinary creations confidently.

4.4 Challenges and Concerns during the Generative Process

4.4.1 Lack of Multi-modal Input for the Interactions and documentation. The current process for documenting ICH culinary practices encounters significant limitations due to the inability to capture multi-modal aspects, such as physical actions and sensory experiences. For example, participant P2 faced challenges when describing the dimensions and preparation method of Chinese sausages. He found it difficult to articulate the precise size and technique using words alone; instead, he resorted to hand gestures to convey how the sausage should be sliced. This reliance on non-verbal communication underscores the inadequacy of textual descriptions in capturing the complexities involved in culinary processes. Furthermore, P2 struggled to articulate the specific state of rice cooking in claypot rice preparation. He explained, *"At the beginning, the rice needs time to cook; it requires time to roast. It's not a matter of cooking it and having it done immediately. The process seems to destroy the ingredients; it's too intense and damages them. It's a slow simmer. At one point, when the charcoal isn't burning, you have to add more charcoal."* Moreover, almost all participants expressed that flavors are intrinsically tied to the experience of cooking and cannot be generated by AI. While visually similar dishes may look alike, they can yield completely different tastes.

4.4.2 Concerns of Balancing Tradition and Innovation. Participants raised several significant concerns about the use of AI in documenting ICH cuisine. One major issue highlighted was the potential for cultural bias in the recipes generated by AI. Participant P12 expressed deep apprehension about this, stating, *"I worry that the recipes generated by AI often reflect a consensus on what is considered delicious, while our family's traditional methods might be overlooked, leading to the loss of our heritage."* From her point of view, this concern underscores a fear that AI could prioritize mainstream culinary trends over the unique, diverse practices that form the backbone of personal and familial cooking traditions. Another critical concern revolved around AI's limitations in capturing authentic human emotions and experiences associated with cooking. Participant P1 articulated this sentiment by noting, *"AI cannot present the things we find truly delicious. It's hard to explain; there's an emotional difference that AI just doesn't convey."* P10 echoed this view, stating, *"AI struggles to represent our years of experience."* He expressed that AI lacks the capability to encompass the depth of knowledge, techniques, and emotional connections that seasoned cooks have cultivated throughout their lives.

5 Discussion

5.1 Document ICH Food In Diverse Scenarios with GenAI

Building upon the foundational work of Fu et al.[34], who demonstrated the efficacy of using GenAI to document and visualize tangible Cultural Heritage artifacts for exhibitions, this study extends the scope of CH documentation to encompass ICH, with a particular focus on culinary practices. They primarily concentrated on the preservation and digital representation of physical CH items, emphasizing accuracy and fidelity in the visual reconstruction of artifacts. In contrast, our research delves into the more fluid and

dynamic nature of ICH, where culinary processes are transmitted through community interactions and improvisational methods.

A key distinction emerging from our findings is the ability to document and explore alternative scenarios within ICH using GenAI. Unlike Fu et al.’s approach, which emphasized the faithful recording of existing artifacts, our methodology facilitates the visualization of contingent cooking practices—such as substituting ingredients, altering techniques, and adapting workflows—that practitioners employ in varying contexts. This capacity to capture and reflect upon alternative scenarios not only aids in preserving the adaptability inherent in ICH but also stimulates creative outcomes among participants.

For instance, participants were able to envision and experiment with ingredient substitutions (e.g., replacing bak choy with cabbage) and alternative cooking methods, leading to innovative dish variations that transcend traditional forms. This aligns with the perspectives of Liu et al. [65], who highlight the role of digital technologies in preserving ICH by making it accessible and adaptable to contemporary contexts. Liu et al. emphasized that digital tools should not only document but also support the evolution of ICH, allowing for both continuity and innovation. Our findings resonate with this viewpoint, demonstrating that GenAI can serve as a catalyst for both preserving traditional culinary methods and inspiring new culinary expressions.

Moreover, our findings contribute to the discourse on HFI by demonstrating how AI-mediated visualization tools can deepen practitioners’ understanding of their culinary heritage. As highlighted by Comber et al. [17], HFI research aims to explore and enhance daily food experiences through technological interventions. Our study complements this by showing that GenAI can facilitate reflective practices and creative experimentation, thereby enriching the culinary experiences and practices of ICH practitioners.

5.2 The limitations of GenAI in ICH

Our study highlights several limitations of GenAI in documenting and visualizing ICH, consistent with previous research findings [34, 66, 75], which noted AI-generated inaccuracies in representing traditional cooking methodologies. Participants in our study frequently observed inaccuracies in ingredient and tool representations, reflecting GenAI’s challenges in accurately capturing non-Western culinary elements due to biases in training datasets.

Cultural bias is another significant concern. Previous researchers discussed cultural biases in GenAI that can lead to the prioritization of mainstream culinary trends over unique family traditions and regional diversity [85]. Such bias poses a threat to preserving and transmitting diverse cultural nuances.

Additionally, GenAI struggles to capture the emotional and experiential dimensions of traditional cooking, often failing to convey the deep knowledge and emotional connections inherent in these practices. These may be due to the lack of the multi-modal interaction which is necessary to fully capture and document complex culinary techniques and sensory experiences. Current text and image outputs are often insufficient to convey the detailed gestures and sensory elements involved in authentic cooking processes.

While GenAI holds promise for prompting cultural reflection and innovation, these limitations in representation accuracy, cultural bias, and interaction hinder its effectiveness.

5.3 Implications

The integration of GenAI into the documentation of ICH culinary practices offers several implications for both practitioners and the design of future AI systems. Firstly, our findings suggest that GenAI can serve as a valuable tool for enhancing reflective practices among culinary practitioners, enabling them to visualize and experiment with alternative cooking scenarios. This capability can foster a deeper understanding of traditional methods while simultaneously encouraging innovation and adaptation.

However, to harness the full potential of GenAI in cultural preservation, it is crucial to establish boundaries that balance creativity with authenticity. Future AI systems should incorporate culturally specific datasets and allow for user-driven customization to ensure that generated content aligns closely with traditional practices. Additionally, incorporating feedback mechanisms where practitioners can iteratively refine and correct AI outputs can enhance the accuracy and cultural relevance of the documentation.

Moreover, the ability of GenAI to stimulate creative outcomes highlights the importance of designing AI tools that not only preserve but also enrich cultural practices. By facilitating the exploration of alternative methodologies and ingredient substitutions, GenAI can inspire practitioners to evolve their culinary traditions in meaningful ways. Designing AI applications with features that support both preservation and innovation can lead to more dynamic and resilient forms of ICH documentation.

Future work may incorporate GenAI and digital processes to reconstruct physical processes in culinary forms with the help of robotics [12, 13, 28, 55, 61] and avatars [33, 103, 104]. Taking advantage of other affordances like smell, haptics [84], voice [83], and touch [56] offer more immersive experiences for representing marginalized communities [29, 58, 101]. These interactions may further communication in communities beyond merely knowledge-based archiving, but rather a spatio-temporal immersive form of communicative action [14].

5.4 Limitations

The demographic composition of the study participants primarily included experienced chefs and long-term practitioners of Chinese cuisine residing abroad. While this provided valuable insights into the adaptive strategies employed in diverse contexts, it also limited the breadth of perspectives, excluding those from different regions. Future research should aim to include a more diverse participant pool to capture a wider range of experiences and approaches to ICH documentation, especially in different culture.

Additionally, the study lacked quantitative measures to assess the perceived usefulness and effectiveness of GenAI-generated documentation. Incorporating such metrics in future studies could provide a more nuanced understanding of how different user groups interact with and benefit from AI-assisted documentation tools. If we could conduct a content analysis of the findings, we may uncover further insights about how the chefs strategically create based on different styles and contingencies that match their expectation.

Furthermore, the contingencies identified during the documentation process were assessed on a case-by-case basis, lacking a standardized framework for categorization and analysis. Developing a more precise taxonomy of contingencies in ICH culinary practices would enhance the ability to systematically capture and compare alternative scenarios across different cultural contexts.

6 Conclusion

This study delved into using GenAI to document alternative scenarios of the Chinese culinary Intangible Cultural Heritage. GenAI proved useful in documenting diverse alternative scenarios in ICH culinary practices, such as ingredient substitutions, cooking for different audiences, and variations in dishes. It also inspired participants to reflect on cuisine - making details, revive cultural memories, explore creative dish variations, and acquire new cooking knowledge. However, GenAI has notable limitations. It often misrepresents ingredients, utensils, cooking stages, and exhibits an unintended aesthetic style. Cultural biases in its training data and the lack of multi-modal input are the main culprits. Moreover, balancing tradition and innovation when using GenAI is challenging, raising concerns about cultural bias and the inability to convey emotional experiences. Our work shows the potentials of GenAI aiding ICH by focusing on its intangible culinary aspect. This process not only preserves the adaptability of ICH but also generates new intangible data, ensuring the continuous evolution and survival of ICH cuisine in the face of modern challenges.

References

- [1] 2009. UNESCO - Text of the Convention for the Safeguarding of the Intangible Cultural Heritage. <https://ich.unesco.org/en/convention>
- [2] 2025. Representative List of National Intangible Cultural Heritage. <https://www.ihchina.cn/project> Accessed: 2025-01-21.
- [3] Mahmoud Al-Sarayreh, Mariza Gomes Reis, Alistair Carr, and Marlon Martins dos Reis. 2023. Inverse design and AI/Deep generative networks in food design: A comprehensive review. *Trends in Food Science & Technology* 138 (Aug. 2023), 215–228. doi:10.1016/j.tifs.2023.06.005
- [4] Marilena Alivizatou-Barakou, Alexandros Kitsikidis, ..., and Nikos Grammalidis. 2017. Intangible Cultural Heritage and New Technologies: Challenges and Opportunities for Cultural Preservation and Development. In *Mixed Reality and Gamification for Cultural Heritage*, Marinos Ioannides, Nadia Magnenat-Thalmann, and George Papagiannakis (Eds.). Springer International Publishing, Cham, 129–158. doi:10.1007/978-3-319-49607-8_5
- [5] Ferran Altarriba Bertran, Jared Duval, Elena Márquez Segura, Laia Turmo Vidal, Yoram Chisik, Marina Juanet Casulleras, Oscar García Pañella, Katherine Isbister, and Danielle Wilde. 2020. Chasing Play Potentials in Food Culture: Learning from Traditions to Inspire Future Human-Food Interaction Design. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference (DIS '20)*. Association for Computing Machinery, New York, NY, USA, 979–991. doi:10.1145/3357236.3395575 TLDR: This pictorial turns to culture and traditions to present an annotated portfolio of play-food potentials, i.e. interesting design qualities and/or interaction mechanisms that could help promote playful and social engagement in food practices..
- [6] Laura Barendregt and Nora S Vaage. 2021. Speculative design as thought experiment. *She Ji: The Journal of Design, Economics, and Innovation* 7, 3 (2021), 374–402.
- [7] Mafkiresseb Kassahun Bekele, Roberto Pierdicca, Emanuele Frontoni, Eva Savina Malinverni, and James Gain. 2018. A Survey of Augmented, Virtual, and Mixed Reality for Cultural Heritage. *J. Comput. Cult. Herit.* 11, 2 (March 2018), 7:1–7:36. doi:10.1145/3145534 TLDR: The article surveys the state-of-the-art in augmented-, virtual-, and mixed-reality systems as a whole and from a cultural heritage perspective and identifies specific application areas in digital cultural heritage and makes suggestions as to which technology is most appropriate in each case..
- [8] Jenny Berkholz, Margarita Esau-Held, Alexander Boden, Gunnar Stevens, and Peter Tolmie. 2023. Becoming an Online Wine Taster: An Ethnographic Study on the Digital Mediation of Taste. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW1 (2023), 26:1–26:26. doi:10.1145/3579459 TLDR: The materials, associated meanings, competences, procedures, and engagements that shaped the performative character of tasting practices are investigated and the implications for the further mediation of taste as a social and democratized phenomenon through online interaction are explored..
- [9] CHIARA BORTOLOTTI. 2007. FROM OBJECTS TO PROCESSES: UNESCO'S 'INTANGIBLE CULTURAL HERITAGE'. *Journal of Museum Ethnography* 19 (2007), 21–33. <http://www.jstor.org/stable/40793837>
- [10] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (Jan. 2006), 77–101. doi:10.1191/1478088706qp0630a
- [11] Tomer Broude. 2015. *A Diet Too Far?: Intangible Cultural Heritage, Cultural Diversity, and Culinary Practices*. Cambridge University Press, 472–493.
- [12] Xiaoyu Chang, Yanheng Li, Sijia Liu, Ling Ma, and Ray LC. 2024. "Sorry to Keep You Waiting": Recovering from Negative Consequences Resulting from Service Robot Unintended Rejection. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI '24)*. Association for Computing Machinery, New York, NY, USA, 96–105. doi:10.1145/3610977.3634959
- [13] Xiaoyu Chang, Fan Zhang, Kexue Fu, Carla Diana, Wendy Ju, and RAY LC. 2025. A Constructed Response: Designing and Choreographing Robot Arm Movements in Collaborative Dance Improvisation. *Proceedings of the ACM on Human-Computer Interaction* 9, CSCW (Oct. 2025), 32. doi:10.48550/arXiv.2505.23090 arXiv:2505.23090 [cs].
- [14] Yihuan Chen, Kexue Fu, Qianyi Chen, Zhicong Lu, and RAY LC. 2025. The Configuration of Space: Probing the Way Social Interaction and Perception are Affected by Task-Specific Spatial Representations in Online Video Communication. In *HCI in Business, Government and Organizations*, Keng Leng Siau and Fiona Fui-Hoon Nah (Eds.). Springer Nature Switzerland, Cham, 113–132. doi:10.1007/978-3-031-92826-0_8
- [15] Yun Chen, Yiwei Wang, Antoni B. Chan, Jixing Li, and Ray LC. 2025. Once More with (the Right) Feeling: How Historical Fiction Writing Processes of Character Design, Plot Outline, and Context Checking Are Affected by Co-Writing with ChatGPT. In *HCI in Business, Government and Organizations*, Keng Leng Siau and Fiona Fui-Hoon Nah (Eds.). Springer Nature Switzerland, Cham, 79–101. doi:10.1007/978-3-031-92823-9_7
- [16] Jaz Hee-jeong Choi, Conor Linehan, Rob Comber, and John McCarthy. 2012. Food for thought: designing for critical reflection on food practices. In *Proceedings of the Designing Interactive Systems Conference (DIS '12)*. Association for Computing Machinery, New York, NY, USA, 793–794. doi:10.1145/2317956.2318077 TLDR: The workshop addresses the opportunities and challenges for the design of digital interactive systems that engage individuals in critical reflection on their everyday food practices – including designing for engagement in more environmentally aware, socially inclusive, and healthier behaviour..
- [17] Rob Comber, Jaz Choi, Jettie Hoonhout, and Kenton O'Hara. 2014. Designing for human–food interaction: An introduction to the special issue on 'food and interaction design'. *International Journal of Human-Computer Studies* 72 (Feb. 2014), 181–184. doi:10.1016/j.ijhcs.2013.09.001
- [18] Rob Comber, Eva Ganglbauer, Jaz Hee-jeong Choi, Jettie Hoonhout, Yvonne Rogers, Kenton O'Hara, and Julie Maitland. 2012. Food and interaction design: designing for food in everyday life. In *CHI '12 Extended Abstracts on Human Factors in Computing Systems (CHI EA '12)*. Association for Computing Machinery, New York, NY, USA, 2767–2770. doi:10.1145/2212776.2212716 TLDR: This workshop brings together a community of researchers and practitioners in human-food interaction to attend to the practical and theoretical difficulties in designing for human- food interactions in everyday life..
- [19] Paul Coulton, Dan Burnett, and Adrian Gradinar. 2016. Games as speculative design: allowing players to consider alternate presents and plausible features. (2016).
- [20] Alexandra Covaci, Khawla Alhasan, Mayank Loonker, Bernardine Farrell, Luma Tabbaa, Sophia Ppali, and Chee Siang Ang. 2023. No Pie in the (Digital) Sky: Co-Imagining the Food Metaverse. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–17. doi:10.1145/3544548.3581305 TLDR: This work aims to provide designers with inspirations for building a metaverse that provides inclusive opportunities for the future of food; helps re-discover the forgotten or lost knowledge about food; facilitates the exploration, excitement and joy of eating; and reinvigorates the ways that food can soothe and heal..
- [21] Chenyang Cui, Youbing Zhao, and Lejing Wang. 2021. Protection and dissemination of Chinese intangible cultural heritage based on digital games. *International Communication of Chinese Culture* 8, 4 (Dec. 2021), 483–491. doi:10.1007/s40636-021-00237-x TLDR: It is proposed that digital games should be used as a new media in the protection and dissemination of Chinese intangible cultural heritage..
- [22] Lucio Tommaso De Paolis, Sofia Chiarello, Carola Gatto, Silvia Liaci, and Valerio De Luca. 2022. Virtual reality for the enhancement of cultural tangible and intangible heritage: The case study of the Castle of Corsano. *Digital Applications in Archaeology and Cultural Heritage* 27 (Dec. 2022), e00238. doi:10.1016/j.daach.2022.e00238

- [23] Angus Deaton and Christina Paxson. 1998. Economies of scale, household size, and the demand for food. *Journal of political economy* 106, 5 (1998), 897–930.
- [24] Jialin Deng, Yan Wang, Carlos Velasco, Ferran Altarriba Altarriba Bertran, Rob Comber, Marianna Obrist, Katherine Isbister, Charles Spence, and Florian 'Floyd' Mueller. 2021. The Future of Human-Food Interaction. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems (CHI EA '21)*. Association for Computing Machinery, New York, NY, USA, 1–6. doi:10.1145/3411763.3441312 TLDR: This workshop aims to explore the possibility of bringing practitioners, researchers and theorists together to discuss the future of human-food interaction with a particular highlight on the design of experiential aspects of human-food interactions beyond the instrumental..
- [25] Jialin Deng, Humphrey Yang, Aryan Saini, Urs Dominic Gaudenz, Lining Yao, Patrick Olivier, and Florian 'Floyd' Mueller. 2023. Dancing Delicacies: Designing Computational Food for Dynamic Dining Trajectories. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23)*. Association for Computing Machinery, New York, NY, USA, 244–262. doi:10.1145/3563657.3596021 TLDR: This paper explores the design of food as a material realization of computation through a material-centered approach by designing a computational food system called "Dancing Delicacies", which enables food items to be "programmed" and "reconfigured" within dynamic trajectories..
- [26] Kira A. Dietz. 2017. Preserving Family Recipes: How to Save and Celebrate Your Food Traditions. *The American Archivist* 80, 2 (Sept. 2017), 470–473. doi:10.17723/0360-9081-80.2.470
- [27] Kosmas Dimitropoulos, Sotiris Manitsaris, Filareti Tsalakanidou, Spiros Nikolopoulos, Bruce Denby, Samer Al Kork, Lise Crevier-Buchman, Claire Pillot-Loiseau, Martine Adda-Decker, Stephane Dupont, Joelle Tilmanne, Michela Ott, Marilena Alivizatos, Erdal Yilmaz, Leontios Hadjileontiadis, Vassilios Charisis, Olivier Deroo, Athanasios Manitsaris, Ioannis Kompatsiaris, and Nikos Grammalidis. 2024. Capturing the Intangible - An Introduction to the I-Treasures Project. 773–781. <https://www.scitepress.org/Link.aspx?doi=10.5220/0004871607730781>
- [28] Kaixu Dong, Zhiyuan Zhang, Xiaoyu Chang, Pakpong Chirarattananon, and RAY LC. 2024. Dances with Drones: Spatial Matching and Perceived Agency in Improvised Movements with Drone and Human Partners. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–16. doi:10.1145/3613904.3642345
- [29] Nianhang Du, Linzhi Wang, Yiran Li, and RAY LC. 2025. Designing for the Invisible: Raising Awareness for the Experience of the Homeless in an LLM-based Serious Game.. In *International Conference on Information Technology for Social Good*. ACM, Antwerp, Belgium, 9. doi:10.1145/3748699.3749822
- [30] Xianzhe Fan, Zihan Wu, Chun Yu, Fenggui Rao, Weinan Shi, and Teng Tu. 2024. ContextCam: Bridging Context Awareness with Creative Human-AI Image Co-Creation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–17. doi:10.1145/3613904.3642129 TLDR: ContextCam is introduced, a novel human-AI image co-creation system that integrates context awareness with mainstream AIGC technologies like Stable Diffusion and leverages Large Language Model-based (LLM) multi-agents to co-create images with the user.
- [31] Valerie J. Frey. 2015. *Preserving Family Recipes: How to Save and Celebrate Your Food Traditions*. University of Georgia Press. Google-Books-ID: CBCiC-gAAQBAJ.
- [32] Jiaying Fu, Yiyang Lu, Zehua Yang, Fiona Nah, and RAY LC. 2025. Cracking Aegis: An Adversarial LLM-based Game for Raising Awareness of Vulnerabilities in Privacy Protection. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference (DIS '25)*. Association for Computing Machinery, New York, NY, USA, 639–662. doi:10.1145/3715336.3735812
- [33] Kexue Fu, Yixin Chen, Jiaxun Cao, Xin Tong, and RAY LC. 2023. "I Am a Mirror Dweller": Probing the Unique Strategies Users Take to Communicate in the Context of Mirrors in Social Virtual Reality. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–19. doi:10.1145/3544548.3581464
- [34] Kexue Fu, Ruishan Wu, Yuying Tang, Yixin Chen, Bowen Liu, and Ray Lc. 2024. "Being Eroded, Piece by Piece": Enhancing Engagement and Storytelling in Cultural Heritage Dissemination by Exhibiting GenAI Co-Creation Artifacts. In *Designing Interactive Systems Conference*. ACM, IT University of Copenhagen Denmark, 2833–2850. doi:10.1145/3643834.3660711
- [35] Tom Gayler, Corina Sas, and Vaiva Kalnikaitė. 2021. Sensory Probes: An Exploratory Design Research Method for Human-Food Interaction. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference (DIS '21)*. Association for Computing Machinery, New York, NY, USA, 666–682. doi:10.1145/3461778.3462013 TLDR: The design and evaluation of Sensory probes are presented, a novel, exploratory design research method aimed to sensitize participants towards their food experiences, and reflections on the value of these probes for enabling novel perspectives on food experiences are provided..
- [36] Qi Gong, Ximing Shen, Ziyou Yin, Yaning Li, and RAY LC. 2025. "If I were in Space": Understanding and Adapting to Social Isolation through Designing Collaborative Storytelling. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference (DIS '25)*. Association for Computing Machinery, New York, NY, USA, 1455–1482. doi:10.1145/3715336.3735846
- [37] A. Grammatikopoulou, S. Laraba, O. Sahbenderoglu, K. Dimitropoulos, S. Douka, and N. Grammalidis. 2019. An adaptive framework for the creation of exergames for intangible cultural heritage (ICH) education. *Journal of Computers in Education* 6, 3 (Sept. 2019), 417–450. doi:10.1007/s40692-018-0115-z
- [38] Yuanning Han, Ziyi Qiu, Jiale Cheng, and RAY LC. 2024. When Teams Embrace AI: Human Collaboration Strategies in Generative Prompting in a Creative Design Task. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3613904.3642133
- [39] Sarah Harmon. 2024. Can AI Help Inspire Healthy Eating? Exploring the Potential of Generated Motivational Texts and Images Related to Healthy Food Choices. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (Honolulu, HI, USA) (CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, Article 67, 5 pages. doi:10.1145/3613905.3650953
- [40] Douglas Harper. 1986. Meaning and Work: A Study in Photo Elicitation. *Current Sociology* 34, 3 (Sept. 1986), 24–46. doi:10.1177/001139286034003006
- [41] Douglas Harper. 2002. Talking about pictures: A case for photo elicitation. *Visual Studies* 17, 1 (April 2002), 13–26. doi:10.1080/14725860220137345 Publisher: Routledge _eprint: <https://doi.org/10.1080/14725860220137345>.
- [42] Zhiting He, Jiayi Su, Li Chen, Tianqi Wang, and RAY LC. 2025. "I Recall the Past": Exploring How People Collaborate with Generative AI to Create Cultural Heritage Narratives. *Proceedings of the ACM on Human-Computer Interaction* 9, CSCW 108 (Oct. 2025), 30. doi:10.1145/3711006
- [43] Yuyao Heng, Yingman Chen, and Zihan Gao. 2024. Echoes of Antiquity: An Interactive Installation for Guqin Cultural Heritage Using Mid-Air interaction and Generative AI. In *SIGGRAPH Asia 2024 Posters (SA '24)*. Association for Computing Machinery, New York, NY, USA, 1–2. doi:10.1145/3681756.3697970
- [44] Jenny L. Herman and Raúl Matta. 2024. Food and UNESCO's Intangible Cultural Heritage Program. In *Oxford Research Encyclopedia of Food Studies*. Oxford University Press. doi:10.1093/acrefore/9780197762530.013.145
- [45] Yumeng Hou, Sarah Kenderdine, Davide Picca, Mattia Egloff, and Alessandro Adamou. 2022. Digitizing Intangible Cultural Heritage Embodied: State of the Art. *J. Comput. Cult. Herit.* 15, 3 (Sept. 2022), 55:1–55:20. doi:10.1145/3494837 TLDR: This article outlines state-of-the-art models, projects, and technical practices that have advanced the digitization lifecycle for ICH resources and aims at identifying the advancements and gaps in the existing conventions, and to envision opportunities for transmitting embodied knowledge in intangible heritage..
- [46] Jingfei Huang, Yan Zhang, Jingyi Ge, Yaning Li, and Ray LC. 2025. The House of the Impossible Gables: Player Engagement and Spatial Perception of Physically Impossible Spaces in Social VR. In *HCI in Business, Government and Organizations*, Keng Leng Siau and Fiona Fui-Hoon Nah (Eds.). Springer Nature Switzerland, Cham, 153–171. doi:10.1007/978-3-031-92826-0_10
- [47] Yi Ji, Tieming Fu, Peng Tan, and Xiaohong Sun. 2019. Research on learning experience of intangible cultural heritage workshop from the perspective of innovation. In *Proceedings of the Seventh International Symposium of Chinese CHI (Chinese CHI '19)*. Association for Computing Machinery, New York, NY, USA, 109–112. doi:10.1145/3332169.3332260
- [48] Rohit Ashok Khot and Florian Mueller. 2019. Human-Food Interaction. *Foundations and Trends® in Human-Computer Interaction* 12, 4 (Aug. 2019), 238–415. doi:10.1561/1100000074 Publisher: Now Publishers, Inc. TLDR: An overview of the existing research in this space is offered, illustrating how HFI builds on recent trends within HCI, and a set of opportunities that emphasize particular features the technology, especially hardware, has yet to offer to drive the human-food interaction field forward are articulated..
- [49] Hyung-Kwon Ko, Gwanmo Park, Hyeon Jeon, Jaemin Jo, Juho Kim, and Jinwook Seo. 2023. Large-scale text-to-image generation models for visual artists' creative works. In *Proceedings of the 28th International Conference on Intelligent User Interfaces*. 919–933.
- [50] Khun Eng Kuah and Zhaohui Liu (Eds.). 2016. *Intangible Cultural Heritage in Contemporary China: The participation of local communities*. Routledge, London. doi:10.4324/9781315628974
- [51] Ray LC. 2023. Human Enough: A Space for Reconstructions of AI visions in Speculative Climate Futures. In *Proceedings of the 15th Conference on Creativity and Cognition (C&C '23)*. Association for Computing Machinery, New York, NY, USA, 217–222. doi:10.1145/3591196.3593341
- [52] Ray LC. 2023. Together Enough: Collaborative Constructions of Adaptations to Climate Futures. In *Companion Publication of the 2023 ACM Designing Interactive Systems Conference (DIS '23 Companion)*. Association for Computing Machinery, New York, NY, USA, 55–59. doi:10.1145/3563703.3596805
- [53] RAY LC. 2024. THE PRESENT IS IN THE FUTURE: Participatory Generative AI Co-Created Visions as Intangible Cultural Heritage. In *Proceedings of the 17th International Symposium on Visual Information Communication and Interaction (VINCI '24)*. Association for Computing Machinery, New York, NY, USA, 1–2. doi:10.1145/3678698.3687200
- [54] Ray LC, Sijia Liu, Latisha Besariani Hendra, and Kexue Fu. 2024. Time Enough: Generative AI Visions of Climate Change as Cave Paintings of the Future. In

- Proceedings of the 16th Conference on Creativity & Cognition (C&C '24)*. Association for Computing Machinery, New York, NY, USA, 608–613. doi:10.1145/3635636.3672190
- [55] RAY LC, Sijia Liu, and Qiaosheng Lyu. 2023. IN/ACTIVE: A Distance-Technology-Mediated Stage for Performer-Audience Telepresence and Environmental Control. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*. Association for Computing Machinery, New York, NY, USA, 6989–6997. doi:10.1145/3581783.3613791
- [56] RAY LC, Sihuang Man, Xiying Bao, Jinhan Wan, Bo Wen, and Zijing Song. 2023. "Contradiction pushes me to improvise": Performer Expressivity and Engagement in Distanced Movement Performance Paradigms. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW2 (Oct. 2023), 333:1–333:26. doi:10.1145/3610182
- [57] Ray LC and Yuying Tang. 2024. Speculative Design with Generative AI: Applying Stable Diffusion and ChatGPT to imagining climate change futures. In *Proceedings of the 11th International Conference on Digital and Interactive Arts (ARTECH '23)*. Association for Computing Machinery, New York, NY, USA, 1–8. doi:10.1145/3632776.3632827
- [58] Ray LC, Anika Ullah, and Fabeha Monir. 2022. A Case for Play: Immersive Storytelling of Rohingya Refugee Experience. In *International Symposium on Electronic Art*. Barcelona, Spain. <https://isea2022.isea-international.org/event/full-paper-a-case-for-play-immersive-storytelling-of-rohingya-refugee-experience/>
- [59] Kai-Sean Lee. 2023. Cooking up food memories: A taste of intangible cultural heritage. *Journal of Hospitality and Tourism Management* 54 (March 2023), 1–9. doi:10.1016/j.jhtm.2022.11.005
- [60] Meng Li, Yaning Li, Chenyang He, Houxiang Wang, Jiaqian Zhong, Shengxin Jiang, Mingtao He, Zhining Qiao, Jiawang Chen, Yuan Yin, Ray LC, Ji Han, Zengyao Yang, and Mohammad Shidujaman. 2024. Generative AI for Sustainable Design: A Case Study in Design Education Practices. In *Human-Computer Interaction*, Masaaki Kurosu and Ayako Hashizume (Eds.). Springer Nature Switzerland, Cham, 59–78. doi:10.1007/978-3-031-60441-6_5
- [61] Yanheng Li, Lin Luoying, Xinyan Li, Yaxuan Mao, and Ray LC. 2023. "Nice to meet you!": Expressing Emotions with Movement Gestures and Textual Content in Automatic Handwriting Robots. In *Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI '23)*. Association for Computing Machinery, New York, NY, USA, 71–75. doi:10.1145/3568294.3580045
- [62] Yaning Li, Sirui Wang, Ximu Sun, Lei Yang, Tingting Zhu, Yuexi Chen, Ke Zhao, Yiqing Zhao, Meng Li, and Ray LC. 2025. Reality as Imagined: Design and Evaluation of a TeleAbsence-Driven Extended Reality Experience for (Re) Interpreting Urban Cultural Heritage Narratives Across Time. *International Journal of Human-Computer Interaction* 0, 0 (2025), 1–30. doi:10.1080/10447318.2025.2554296 Publisher: Taylor & Francis _eprint: <https://www.tandfonline.com/doi/pdf/10.1080/10447318.2025.2554296>.
- [63] Martin V. A. Lindrup, EunJeong Cheon, Mikael B. Skov, and Dimitrios Raptis. 2021. One Byte at a Time: Insights about Meaningful Data for Sustainable Food Consumption Practices. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference (DIS '21)*. Association for Computing Machinery, New York, NY, USA, 683–696. doi:10.1145/3461778.3462121 TLD: The results show the way in which the current food data is challenging the understanding of its environmental impacts, which concern data of availability, data representations, and data cultures..
- [64] Long Ling, Xinyi Chen, Ruoyu Wen, Toby Jia-Jun Li, and RAY LC. 2024. Sketchar: Supporting Character Design and Illustration Prototyping Using Generative AI. *Proc. ACM Hum.-Comput. Interact.* 8, CHI PLAY (Oct. 2024), 337:1–337:28. doi:10.1145/3677102
- [65] Guanhong Liu, Xianghua(Sharon) Ding, Jinghe Cai, Weiyun Wang, Xinyue Wang, Yuting Diao, Jin Chen, Tianyu Yu, Haiqing Xu, and Haipeng Mi. 2023. Digital Making for Inheritance and Enlivening Intangible Cultural Heritage: A Case of Hair Monkey Handicrafts. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, Hamburg Germany, 1–17. doi:10.1145/3544548.3581539
- [66] Sijia Liu, Xiaoke Zeng, Fengyihan Wu, Shu Ye, Bowen Liu, Sydney Cheung, Richard William Allen, and RAY LC. 2025. "Salt is the Soul of Hakka Baked Chicken": Reimagining Traditional Chinese Culinary ICH for Modern Contexts Without Losing Tradition. In *Creativity and Cognition (C&C '25)*. Association for Computing Machinery, New York, NY, USA, 11. doi:10.1145/3698061.3726917
- [67] Zhicong Lu. 2019. Live streaming in China for sharing knowledge and promoting intangible cultural heritage. *interactions* 27, 1 (Dec. 2019), 58–63. doi:10.1145/3373145
- [68] Zhicong Lu, Michelle Annett, Mingming Fan, and Daniel Wigdor. 2019. "I feel it is my responsibility to stream": Streaming and Engaging with Intangible Cultural Heritage through Livestreaming. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3290605.3300459 TLD: It was found that ICH streamers had altruistic motivations and engaged with viewers using multiple modalities beyond livestreams, while non-live curated videos attracted attention from a broader audience and assisted in the archiving of knowledge..
- [69] Zhicong Lu, Haijun Xia, Seongkook Heo, and Daniel Wigdor. 2018. You Watch, You Give, and You Engage: A Study of Live Streaming Practices in China. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3173574.3174040
- [70] Lucas M. Silva and Daniel A. Epstein. 2021. Investigating Preferred Food Description Practices in Digital Food Journaling. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference (DIS '21)*. Association for Computing Machinery, New York, NY, USA, 589–605. doi:10.1145/3461778.3462145 TLD: A food journal prototype supporting varied devices and input modalities was deployed, which 15 participants used to journal 1008 food logs over two weeks, and participants had diverse strategies for indicating what and how much they ate..
- [71] Ling Ma, Mingyao Pan, Vince Siu, Xiaoyu Chang, Jussi Jolopainen, Jixing Li, and Ray LC. 2025. Follow My Logic: Generative AI Workflows in Designing for Serious Table-Top Games. In *HCI in Business, Government and Organizations*, Keng Leng Siau and Fiona Fui-Hoon Nah (Eds.). Springer Nature Switzerland, Cham, 153–172. doi:10.1007/978-3-031-92823-9_11
- [72] Teng-Fei Ma, Chang-Wei Chai, and Tseng-Wei Chao. 2022. On the Study of the Sustainable Development of Intangible Cultural Heritage of Indigenous Peoples' Diets—Take the Protection of Geographical Indications as an Example. *Sustainability* 14, 19 (2022). doi:10.3390/su141912803
- [73] Esquire Magazine. 2023. Taste of Intangible Cultural Heritage. *NetEase Subscription* (2023). <https://www.163.com/dy/article/IK3Cl6AS0518DG2J.html>
- [74] Atefeh Mahdavi Goloujeh, Anne Sullivan, and Brian Magerko. 2024. Is It AI or Is It Me? Understanding Users' Prompt Journey with Text-to-Image Generative AI Tools. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–13. doi:10.1145/3613904.3642861 TLD: The prompt journey as an individual yet a social experience is discussed and opportunities for aligning text-to-image AI tools and users' intents are highlighted..
- [75] Marty Miller, Zhiyuan Zhang, Jing Liao, and RAY LC. 2025. Eliciting (Immersive) Intangibles: GenAI-Supported Collaborative Visual Narration In a Physically Immersive Space. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference (DIS '25)*. Association for Computing Machinery, New York, NY, USA, 3313–3325. doi:10.1145/3715336.3735682
- [76] Fnu Mohbat and Mohammed J. Zaki. 2024. LLaVA-Chef: A Multi-modal Generative Model for Food Recipes. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management* (Boise, ID, USA) (CIKM '24). Association for Computing Machinery, New York, NY, USA, 1711–1721. doi:10.1145/3627673.3679562
- [77] Michela Mortara, Chiara Eva Catalano, Francesco Bellotti, Giusy Fiucci, Minica Houry-Panchetti, and Panagiotis Petridis. 2014. Learning cultural heritage by serious games. *Journal of Cultural Heritage* 15, 3 (May 2014), 318–325. doi:10.1016/j.culher.2013.04.004
- [78] Adebayo Oyefunke Olayemi. 2012. Effects of family size on household food security in Osun State, Nigeria. *Asian journal of agriculture and rural development* 2, 2 (2012), 136–141.
- [79] Thomas A. Reuter. 2022. Endangered Food Systems: Agriculture, Nutrition and Cultural Heritage in Bali, Indonesia. *Unisia* (Aug. 2022), 147–160. doi:10.20885/unisia.vol40.iss1.art7
- [80] Marco Romagnoli. 2019. Gastronomic heritage elements at UNESCO: problems, reflections on and interpretations of a new heritage category. *International Journal of Intangible Heritage* 14 (Dec. 2019).
- [81] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. doi:10.48550/arXiv.2112.10752 arXiv:2112.10752 [cs].
- [82] Elmedin Selmanović, Selma Rizvic, Carlo Harvey, Dusanka Boskovic, Vedad Hulusic, Malek Chahin, and Sanda Sljivo. 2020. Improving Accessibility to Intangible Cultural Heritage Preservation Using Virtual Reality. *J. Comput. Cult. Herit.* 13, 2 (May 2020), 13:1–13:19. doi:10.1145/3377143
- [83] Chenxinran Shen, Yan Xu, RAY LC, and Zhicong Lu. 2024. Seeking Soulmate via Voice: Understanding Promises and Challenges of Online Synchronized Voice-Based Mobile Dating. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3613904.3642860
- [84] Ximing Shen, Xuan Li, Youichi Kamiyama, Danny Hynds, Giulia Barbareschi, RAY LC, Sohei Wakisaka, Arata Horie, and Kouta Minamizawa. 2025. "It's Like Being On Stage": Conveying Dancers' Expressiveness Through A Haptic-Installed Contemporary Dance Performance. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 18. doi:10.1145/3706598.3713321
- [85] Jisu Shin, Hoyun Song, Huije Lee, Soyeong Jeong, and Jong Park. 2024. Ask LLMs Directly, "What shapes your bias?": Measuring Social Bias in Large Language Models. In *Findings of the Association for Computational Linguistics: ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 16122–16143. doi:10.18653/v1/2024.findings-acl.954

- [86] M Sivakaran. [n. d.]. Journal of Intangible Cultural Heritage. *Journal of Intangible Cultural Heritage* ([n. d.]), 13.
- [87] Huiyun Tang, Songqi Sung, Xexin Nie, Ang Li, Anastasi Sergeeva, and RAY LC. 2025. Breaking the News: Taking the Roles of Influencer vs Journalist in a LLM-based Game for Raising Misinformation Awareness. *Proc. ACM Hum.-Comput. Interact.* 9, CHI PLAY (Oct. 2025), 6: 42. doi:10.1145/3748600
- [88] Antonia Trichopoulou, Stavroula Soukara, and Effie Vasilopoulou. 2007. Traditional foods: a science and society perspective. *Trends in Food Science & Technology* 18, 8 (2007), 420–427.
- [89] UNESCO. 2003. Convention for the Safeguarding of the Intangible Cultural Heritage 2003. <https://ich.unesco.org/en/convention>
- [90] UNESCO. 2010. Decision of the Intergovernmental Committee: Representative List of the Intangible Cultural Heritage of Humanity. <https://ich.unesco.org/en/decisions>
- [91] UNESCO. 2010. Traditional Mexican cuisine - ancestral, ongoing community culture, the Michoacán paradigm. <https://ich.unesco.org/en/RL/traditional-mexican-cuisine-ancestral-ongoing-community-culture-the-michoacan-paradigm-00400>
- [92] UNESCO. 2021. Success story of promoting traditional foods and safeguarding traditional foodways in Kenya. <https://ich.unesco.org/en/BSP/success-story-of-promoting-traditional-foods-and-safeguarding-traditional-foodways-in-kenya-01409>
- [93] UNESCO. 2023. Practices and meanings associated with the preparation and consumption of ceviche, an expression of Peruvian traditional cuisine. <https://ich.unesco.org/en/RL/practices-and-meanings-associated-with-the-preparation-and-consumption-of-ceviche-an-expression-of-peruvian-traditional-cuisine-01952>
- [94] UNESCO. 2024. Summer farming at fäbod and seter: knowledge, traditions and practices related to the grazing of outlying lands and artisan food production. <https://ich.unesco.org/en/RL/summer-farming-at-fabod-and-seter-knowledge-traditions-and-practices-related-to-the-grazing-of-outlying-lands-and-artisan-food-production-02109>
- [95] Mathias Peter Verheijden and Mathias Funk. 2023. Collaborative Diffusion: Boosting Designerly Co-Creation with Generative AI. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, Hamburg Germany, 1–8. doi:10.1145/3544549.3585680 TLDR: This paper introduces BrainFax, a tool that facilitates generating, editing, and sharing images with AI through a chat bot and online whiteboard, and finds that AI image generation can boost designerly co-creation, yet needs careful embedding into a workflow to leverage inspirational and communicative creations..
- [96] Huanchen Wang, Minzhu Zhao, Wanyang Hu, Yuxin Ma, and Zhicong Lu. 2024. Critical Heritage Studies as a Lens to Understand Short Video Sharing of Intangible Cultural Heritage on Douyin. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–21. doi:10.1145/3613904.3642138
- [97] Yuqi Wang, Sirui Wang, Shiman Zhang, Kexue Fu, Michelle Lui, and RAY LC. 2025. From Temporal to Spatial: Designing Spatialized Interactions with Segmented-audios in Immersive Environments for Active Engagement with Performing Arts Intangible Cultural Heritage. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference (DIS '25)*. Association for Computing Machinery, New York, NY, USA, 3292–3312. doi:10.1145/3715336.3735787
- [98] Philip Weber, Thomas Ludwig, Sabrina Brodesser, and Laura Grönwald. 2021. "It's a Kind of Art!": Understanding Food Influencers as Influential Content Creators. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3411764.3445607 TLDR: A process model is derived that outlines the practices of food influencers as influential content creators within the social media sphere to open up design spaces for interaction researchers and practitioners in the field of HCI..
- [99] Justin D. Weisz, Jessica He, Michael Muller, Gabriela Hoefer, Rachel Miles, and Werner Geyer. 2024. Design Principles for Generative AI Applications. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–22. doi:10.1145/3613904.3642466 TLDR: Six principles for the design of generative AI applications are presented that address unique characteristics of generative AI UX and offer new interpretations and extensions of known issues in the design of AI applications..
- [100] Daijin Yang, Yanpeng Zhou, Zhiyuan Zhang, Toby Jia-Jun Li, and RAY LC. 2022. AI as an Active Writer: Interaction strategies with generated text in human-AI collaborative fiction writing. In *Joint Proceedings of the UI 2022 Workshops: APEX-UI, HAI-GEN, HEALTHI, HUMANIZE, TExSS, SOCIALIZE*. CEUR-WS Team, 56–65. [https://scholars.cityu.edu.hk/en/publications/publication\(d901f5a2-0600-422f-b588-db5a59871961\).html](https://scholars.cityu.edu.hk/en/publications/publication(d901f5a2-0600-422f-b588-db5a59871961).html)
- [101] Hongni Ye, Chaoyu Zhang, Hongsheng Xu, Ray LC, and Xin Tong. 2022. Twilight Rohingya: The Design and Evaluation of Different Navigation Controls in a Refugee VR Environment. In *2022 International Conference on Cyberworlds (CW)*. 142–146. doi:10.1109/CW55638.2022.00032 ISSN: 2642-3596.
- [102] Yuhang Zeng, Yingxuan Shi, Xuehan Huang, Fiona Nah, and RAY LC. 2025. "Ronaldo's a poser!": How the Use of Generative AI Shapes Debates in Online Forums. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 18. doi:10.1145/3706598.3713829
- [103] Fan Zhang, Yun Chen, Xiaoke Zeng, Tianqi Wang, Long Ling, and RAY LC. 2025. "An Image of Ourselves in Our Minds": How College-educated Online Dating Users Construct Profiles for Effective Self Presentation. *Proceedings of the ACM on Human-Computer Interaction* 9, CSCW 031 (Oct. 2025), 32. doi:10.1145/3710929
- [104] Fan Zhang, Molin Li, Xiaoyu Chang, Kexue Fu, Richard William Allen, and RAY LC. 2025. "Becoming My Own Audience": How Dancers React to Avatars Unlike Themselves in Motion Capture-Supported Live Improvisational Performance. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 20. doi:10.1145/3706598.3713390
- [105] Qinshi Zhang, Ruoyu Wen, Latisha Besariani Hendra, Zijian Ding, and RAY LC. 2025. Can AI Prompt Humans? Multimodal Agents Prompt Players' Game Actions and Show Consequences to Raise Sustainability Awareness. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 29. doi:10.1145/3706598.3713661
- [106] Suifang Zhou, Kexue Fu, Huamin Yi, and RAY LC. 2025. RetroChat: Designing for the Preservation of Past Chinese Online Social Experiences. In *Creativity and Cognition (C&C '25)*. Association for Computing Machinery, New York, NY, USA, 19. doi:10.1145/3698061.3726920
- [107] Suifang Zhou, Latisha Hendra, and Ray LC. 2024. Eternagram: Post-Climate Devastation Text Adventure. In *Proceedings of the 17th International Symposium on Visual Information Communication and Interaction (VINCI '24)*. Association for Computing Machinery, New York, NY, USA, 1–2. doi:10.1145/3678698.3687201
- [108] Suifang Zhou, Latisha Besariani Hendra, Qinshi Zhang, Jussi Holopainen, and RAY LC. 2024. Eternagram: Probing Player Attitudes Towards Climate Change Using a ChatGPT-driven Text-based Adventure. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–23. doi:10.1145/3613904.3642850
- [109] Suifang Zhou and Ray LC. 2025. Eternagram: Inspiring Climate Action Through LLM-based Conversational Exploration of a Post-Devastation Climate Future. In *Proceedings of the 7th ACM Conference on Conversational User Interfaces (CUI '25)*. Association for Computing Machinery, New York, NY, USA, 1–7. doi:10.1145/3719160.3735658